1. **Title**

Electricity Demand Forecasting Using HistGradientBoosting and SARIMAX: A Comparative Analysis

2. **Project Overview**

Electricity demand forecasting is vital for ensuring a reliable and efficient power supply. Utility companies and grid operators rely on demand forecasts to support generation planning, resource allocation, and operational decision-making. Reliable forecasts help reduce energy shortages, mitigate operational costs, and enhance the overall stability of the power system.

My objective of this project was to investigate to what extent can tree-based machine learning exploit lagged, rolling and calendar information to improve short-term electricity demand forecasting relative to classical seasonal time-series modelling, using historical electricity demand data. Two different forecasting methods were used and evaluated: the HistGradientBoosting algorithm, a machine learning model, and SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) model, a statistical time-series forecasting technique.

The project was implemented entirely in Python, covering data preprocessing, feature engineering, model development, prediction, visualization, and performance evaluation. The forecasting accuracy of both models was compared using standard evaluation metrics to determine their effectiveness in predicting future electricity demand and to highlight the strengths and limitations of machine learning and statistical approaches for time-series forecasting.

3. Dataset

The dataset used in this project was obtained from Kaggle and contains historical hourly electricity demand records for the PJM West (PJMW) region. PJM Interconnection is a regional transmission organization responsible for coordinating the movement of wholesale electricity across several states in the United States.

3.2 The dataset consists of 114,900 observations and includes the following variables:

3.3. Datetime: The timestamp indicating the date and hour of each electricity demand measurement.

3.4. PJMW_MW: The hourly electricity demand, measured in megawatts (MW), which serves as the target variable for forecasting.

The Datetime variable was converted to a datetime format and used as the time index for the analysis. The dataset was then prepared for forecasting through preprocessing and feature engineering, enabling both the HistGradientBoosting and SARIMAX models to effectively learn temporal patterns in the series.

4. Data Preparation

4.1. Feature engineering

To improve the predictive performance of machine learning model, several feature engineering techniques were applied to capture the temporal characteristics of electricity demand. These included:

Lag features: Previous electricity demand values were created as predictor variables to capture temporal dependencies.

Rolling statistics: Rolling window features, such as moving averages, were generated to represent recent demand trends and smooth short-term fluctuations.

Calendar features: Time-based variables, including the hour of the day, day of the week, month, and other relevant calendar attributes, were extracted from the datetime index to capture daily, weekly, and seasonal demand patterns.

4.2. Data splitting

For the HistGradientBoosting model, the dataset was divided into 80% training data and 20% testing data, allowing the model to learn from historical observations while evaluating its forecasting performance on unseen data.

Due to the higher computational complexity of the SARIMAX model, training was performed using the most recent 10,000 observations rather than the full dataset. This approach significantly reduced computation time while retaining sufficient historical information to model the temporal structure of the electricity demand series effectively.

5. Methodology

Two forecasting approaches were implemented to predict future electricity demand: HistGradientBoosting as a machine learning approach and SARIMAX as a statistical time-series approach. The objective was to compare the predictive performance of these two approaches when applied to the same electricity demand forecasting problem.

5.1. HistGradientBoosting

HistGradientBoosting is a gradient boosting algorithm that builds an ensemble of decision trees sequentially, with each new tree attempting to improve the errors made by the previous trees. The histogram-based implementation improves computational efficiency by grouping continuous feature values into discrete bins.

For this project, the model was trained using the engineered lag, rolling, and calendar features. These features allowed the model to learn relationships between previous electricity demand, recent demand trends, and calendar patterns.

The model was trained on 80% of the observations and evaluated on the remaining 20%. Hyperparameters were selected to obtain a suitable balance between predictive performance and computational efficiency.

5.2. SARIMAX

SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous variables) was used as the statistical forecasting approach. Unlike HistGradientBoosting, SARIMAX explicitly models the temporal structure of the demand series through autoregressive, differencing, moving-average, and seasonal components.

The model was fitted using the most recent 10,000 observations because of the substantially higher computational requirements associated with SARIMAX estimation on a large hourly dataset. The model was then used to generate forecasts for the evaluation period.

5.3. Model Comparison

The two models were evaluated using the same forecasting objective and their predictions were compared against the observed electricity demand. Performance was assessed using standard forecasting error metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

6. Implementation

The complete forecasting workflow was implemented in Python, covering data loading, preprocessing, feature engineering, model development, forecasting, evaluation, and visualization. The implementation was designed as a reproducible workflow in which the same dataset was processed and evaluated systematically for both forecasting approaches.

The main Python libraries used in the project included:

Pandas

NumPy

Scikit-learn

Statsmodels

Matplotlib

The implementation followed the following workflow:

Data Loading → Data Inspection → Datetime Processing → Feature Engineering → Train/Test Preparation → Model Training → Forecasting → Model Evaluation → Visualization → Model Comparison

For HistGradientBoosting model, lagged demand variables, rolling statistics, and calendar-based features were used as predictors. The model was trained on the training portion of the dataset and subsequently used to generate predictions for the test period.

For SARIMAX, the electricity demand time series was prepared as a chronological series and the model was fitted using 10,000 observations. Forecasts were then generated for the corresponding evaluation period.

Finally, the predictions from both models were evaluated using the selected forecasting metrics and visualized against the actual electricity demand. This implementation provided a complete Python-based machine learning and statistical forecasting pipeline, demonstrating skills in Python programming, data manipulation, feature engineering, statistical modeling, machine learning, model evaluation, and data visualization.

7. Model Evaluation

The performance of both forecasting models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2. These metrics were selected to evaluate prediction accuracy from different perspectives.

7.1 HistGradientBoosting Results

The HistGradientBoosting model produced the following results on the 20% test dataset:

Metric	**Result**
MAE	368.55 MW
RMSE	489.13 MW
MAPE	1.18%
R^2	0.99

On average, forecast differs from actual electricity demand by around ±369 MegaWatts (MW). This error is quite small since the demand ranges approximately between 20,000 MW to 30,000 MW from actual vs predicted plot, eliminating concerns about grid operators' risk of insufficient generation (under predict) or wasted generation cost (over predict).

The model's typical error magnitude is about 489 MW. This could be that electricity demand might have sudden spikes; heat waves, cold, holidays, industrial surges, however, the model is stable because a 121 MW difference between MAE and RMSE is not huge.

The results indicate strong predictive performance. The model achieved a MAPE of 1.18%, meaning that its forecasts deviated from the actual electricity demand by approximately 1% on average. The R^2 of 0.99 indicates that the model explained approximately 99% of the variation in the test data, and the model captures almost all patterns.

The relatively low MAE and RMSE further indicate that the model was able to closely track the observed hourly electricity demand.

7.2 SARIMAX Results

The SARIMAX model produced the following results:

Metric	Result
MAE	2,005.62 MW
RMSE	2,482.52 MW
R^2	0.688

The SARIMAX model showed considerably larger forecasting errors than the HistGradientBoosting model. Its MAE was approximately 2,005.62 MW, while the RMSE was 2,482.52 MW. The R^2 value of 0.688 indicates that the model explained approximately 68.8% of the variation in the evaluation data.

7.3 Initial Model Comparison

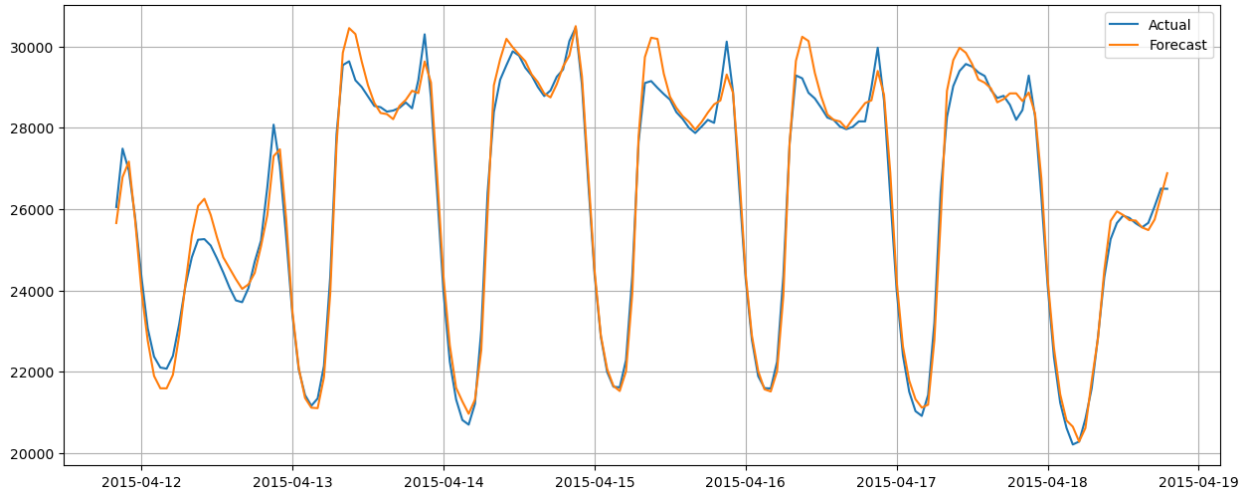
Based on the evaluation metrics, HistGradientBoosting substantially outperformed SARIMAX in this experiment.

The results suggest that the engineered lag, rolling, and calendar features provided HistGradientBoosting with useful information for learning the nonlinear relationships and temporal patterns in electricity demand.

7.2 HistGradientBoosting: Actual vs. Forecast

The figure below compares the observed electricity demand with the values forecast by the HistGradientBoosting model.

Figure 1. Actual vs. Forecast Electricity Demand Using HistGradientBoosting



The forecasted electricity demand closely follows the actual load profile, demonstrating that HistGradientBoosting model effectively learned the dynamics and daily cyclic behavior of electricity consumption. In the night to early morning there is low demand of electricity around 22000 MW, but demand begins to rise steadily throughout the morning to afternoon. Despite reaching approximately 30000 MW at its peak in the afternoon to evening, demand decreases drastically to about 20000 in late night, and the cycle continue the next day. Minor deviations are observed during peak demand periods, showing uncertainty during high volatile periods.

The close alignment between the actual and forecast series is consistent with the model's strong quantitative performance, including an RMSE of 489.13 MW, MAE of 368.55 MW, MAPE of 1.18%, and R^2 of 0.99.

7.3 SARIMAX: Actual vs. Forecast

The second figure compares the observed electricity demand with the forecast generated by the SARIMAX model.

Figure 2. Actual vs. Forecast Electricity Demand Using SARIMAX

The SARIMAX forecasts capture the general temporal structure and recurring patterns in electricity demand, but the forecast series shows larger deviations from the actual observations than the HistGradientBoosting model. This visual observation is consistent with its higher MAE of 2,005.62 MW, RMSE of 2,482.52 MW, and lower R^2 of 0.688.

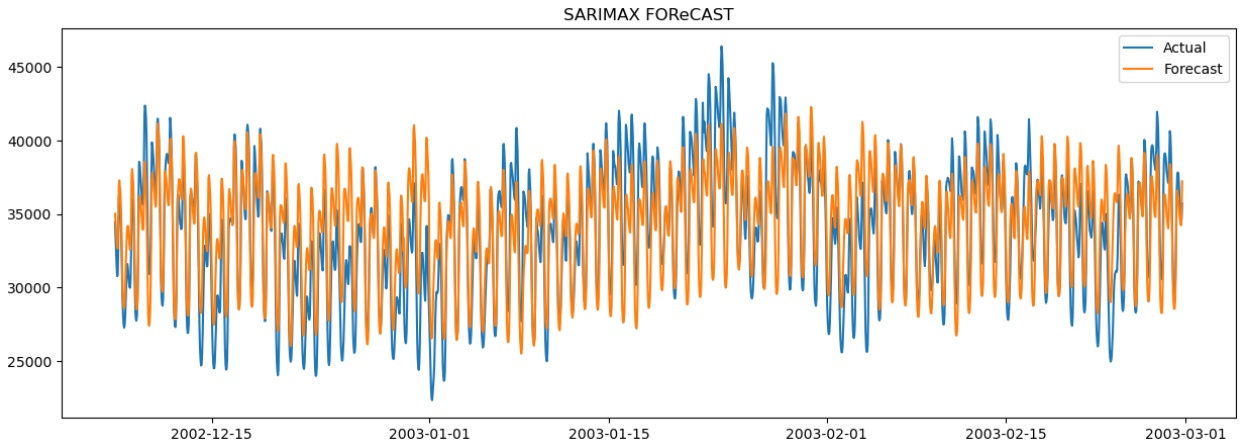


Figure 2 illustrates the actual and predicted electricity demand generated by the SARIMAX model.

7.4. Visual Comparison

The forecast plots provide an additional perspective beyond the numerical evaluation metrics. HistGradientBoosting demonstrates a closer correspondence between actual and predicted demand, particularly in capturing the short-term fluctuations and recurring peaks and troughs in the hourly series. SARIMAX captures the broader temporal pattern but shows larger deviations from the observed demand.

8. Discussion

The results demonstrate a clear difference in the forecasting performance of the two approaches. HistGradientBoosting achieves substantially better predictive accuracy over SARIMAX across the main evaluation metrics. The HistGradientBoosting model achieved an MAE of 368.55 MW, RMSE of 489.13 MW, and R^2 of 0.99, compared with an MAE of 2,005.62 MW, RMSE of 2,482.52 MW, and R^2 of 0.688 for SARIMAX.

The strong performance of HistGradientBoosting can be credited partly to the feature engineering performed during the project. The use of lag features, rolling statistics, and calendar features provided the model with information about previous demand, recent dem

and trends, and recurring hourly and calendar patterns. This enabled the model to learn nonlinear relationships within the electricity demand data.

The SARIMAX model was able to capture the broader temporal and seasonal structure of the electricity demand series, as shown in the forecast visualization. However, its forecasts showed larger deviations from the observed values, particularly when attempting to capture short-term fluctuations. This suggests that the statistical structure specified for SARIMAX was less effective at representing some of the complex patterns present in the dataset.

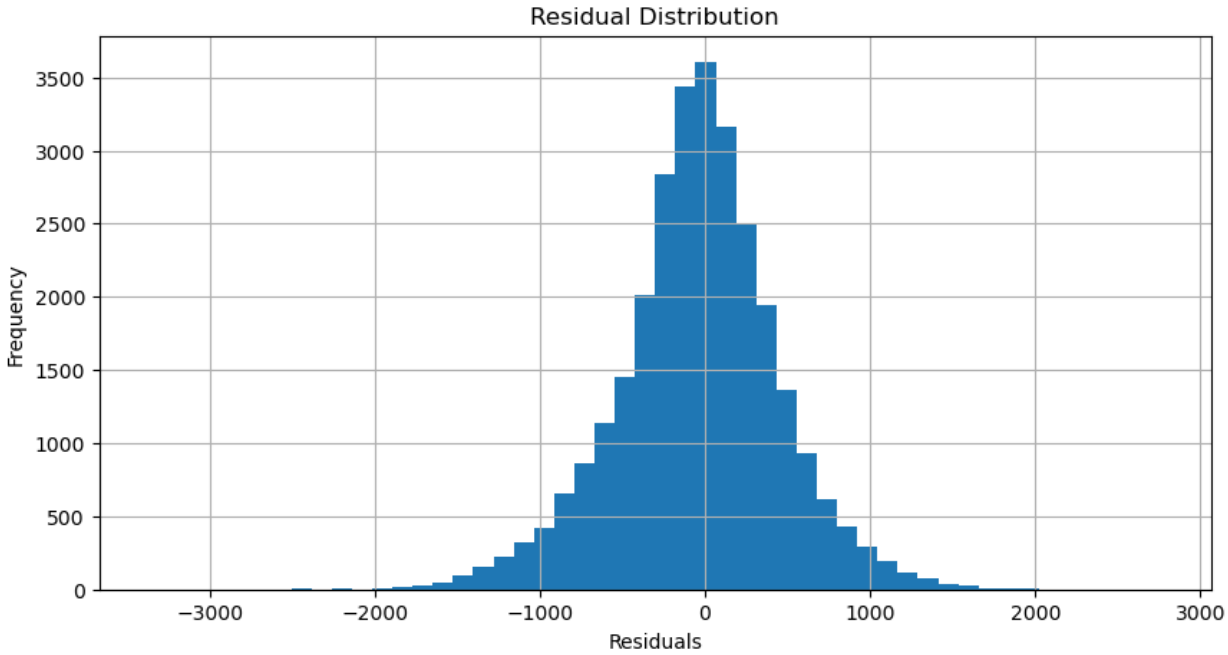
The comparison also highlights an important practical consideration. SARIMAX required substantially greater computational resources when applied to the full dataset, which led to the use of 10,000 observations for its modeling. In contrast, HistGradientBoosting was able to work with the larger dataset more efficiently. Therefore, computational efficiency was another practical advantage of the machine learning approach in this project.

Also the two models were not trained under completely identical conditions. HistGradientBoosting used the full dataset with an 80/20 chronological train-test split, whereas SARIMAX was fitted using 10,000 observations. Consequently, the comparison demonstrates the practical performance of the two approaches in this implementation, but not be interpreted as a perfectly controlled scientific comparison.

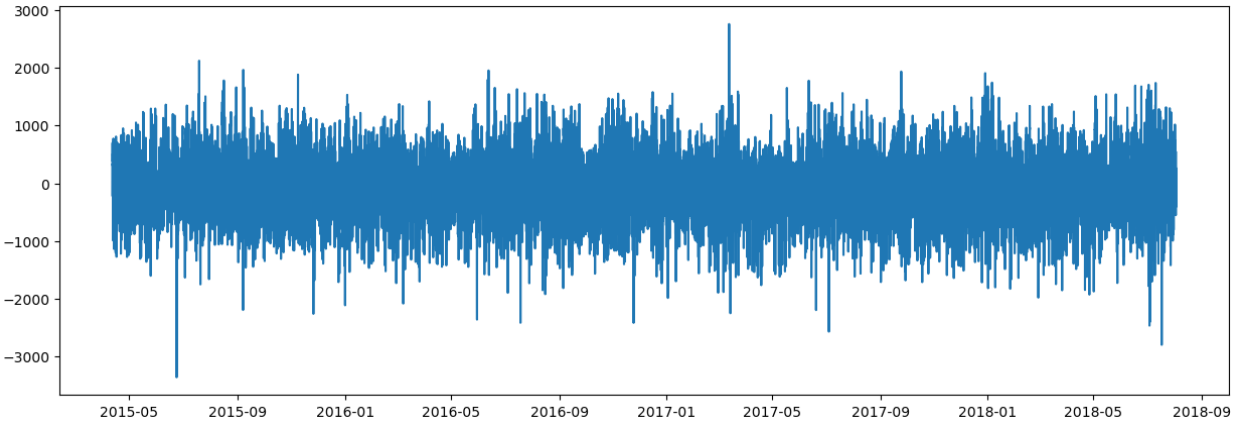
9. Residual Diagno

9.1 **Residual Distribution of Histgradientboosting**

In the figure below, the bell-shaped distribution provides evidence that the residuals have an approximately normal. The residual histogram shows that the errors are concentrated around zero, with the highest frequency occurring near the center of the distribution. The distribution also decreases as the residual moves away from zero, indicating that large predicting errors occur less frequently.

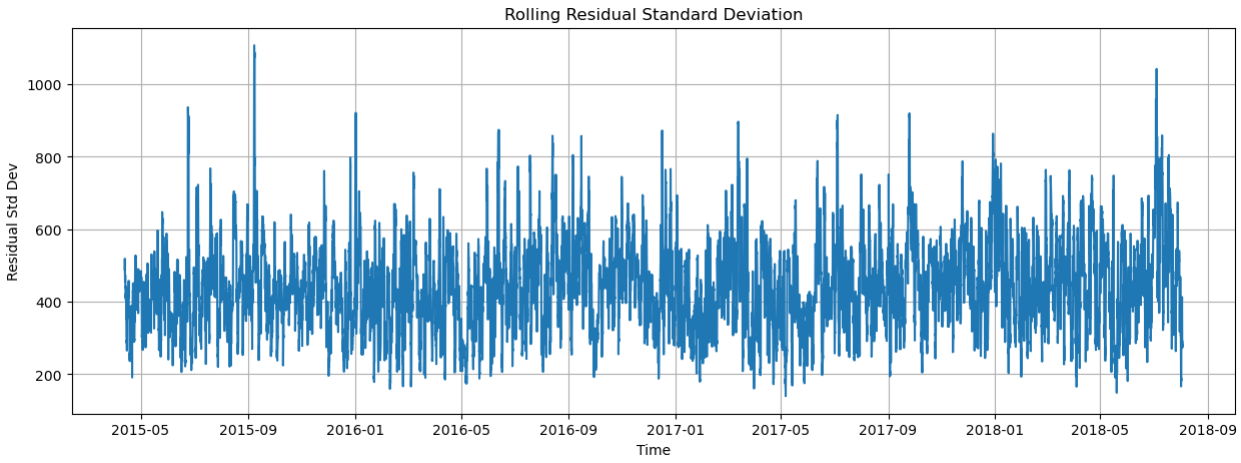


8.2 Residuals over time



Residuals fluctuate around zero and most errors are within approximately -1400 MW to +1400 MW. There are occasional few spikes reaching -3000 MW and +3000 MW. The residuals are randomly distributed around zero, indicating that the HistGradientBoosting model effectively captured the underlying electricity demand dynamics. Occasional large residual spikes correspond to extreme demand events likely influenced by external factors such as weather anomalies or holidays.

8.3 Rolling Residual Variance



Rolling Residual Deviation over time shows that the forecasting errors remain stable across time, residual is mostly stable range, and forecast accuracy is reasonably consistent, yet some period show sudden spikes upward. The Ljung-Box test at lag 24 produced a p-value of approximately 0.00, indicating statistically significant autocorrelation in the residuals.

Although, the HistGradientBoosting model achieved strong predictive performance (R2 = 0.99), residual diagnostics revealed remaining temporal autocorrelation. This demonstrates that high predictive accuracy does not necessarily imply that all underlying temporal dependencies have been captured. The finding also provides a basis for future model improvement through additional temporal and exogenous features.

Overall, the experiment demonstrates that machine learning and statistical forecasting approaches can provide substantially different results when applied to the same time-series forecasting problem. In this case, HistGradientBoosting provided more accurate forecasts and was computationally more practical, while SARIMAX provided a useful statistical benchmark for understanding the temporal structure of electricity demand.

9. POLICY ACTIONS

1. Implement time of use pricing - In other to prevent blackouts, grid distributors may charge higher tariffs during peak hours and lower tariffs during off peak hours to encourage consumers to shift usage away from peak periods, also encourage industries, commercial buildings, and large consumers to reduce usage during peak demands.

2. Renewable energy integration

- Additional power generation such as solar generation and wind generation to support peak demands.
- Households must have backup generators or plant before peak demand occurs.
- Determine when batteries should charge, when stored electricity should discharge, during evening peaks.
- Electricity Management Company must be well equipped to prevent blackouts during extreme weather conditions that may disrupt transformers failures, voltage instability, overloads.